QC STORY PROJECT

Project Title: To Eliminate the Defect of Battery tray bolt loose fitment from Marazzo.

Mahindra and Mahindra Ltd,NASIK_ Plant, _Multimodel TCF.

	Plan	Actual
Start Date	01.04.2018	15.04.2018
Completion Date	11.11.2018	20.11.2018

Current Status: **Completed**

Phase II – QC Story

- Team Information
- Step1: Select the Theme
- Step 2: Justify the choice
- Step 3: Understand the current situation
- Step 4: Select Targets
- Step 5: Analysis (Level 1 Analysis)
- Step 6: Implement corrective measures
- Step 7: Confirm the Effects
- Step 8: Standardize
- Step 9: Summaries & Plan future actions
- Guidelines to user

Team information

No	Description	Name	Designation	Dept	Phone	Email
1	**Team Leader**	Suhas Aher	Deputy Manager	Production	9960615178	Aher.suhas@Mahindra.com
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6	**Member 4**	Sachin Bhamabre	Deputy Manager	MQA	9158960111	Bhamabre.sachin@Mahindra.com
7	**Member 5**	Varun Parakh	Senior Engineer	R & D	7418421841	Parakh.Varun@Mahindra.com
8	**Member 6**	Avinash Mahale	Deputy Manager	SCM	9823996005	Mahale.avinash@Mahindra.com

Step1. Select Subject

▪Brief Description : Elimination of the Metal Battery tray bolt loose fitment

What : Metal battery tray bolt loose

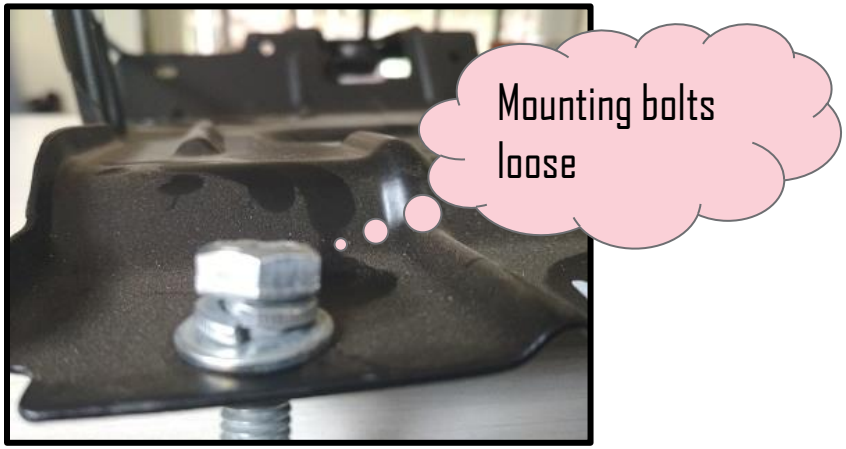
Where : At console stage at Final assembly line.

When : from Jan-18.

Why: Chronic Repeat Concern, Production loss, Quality loss, Cost loss, Deliver loss.

Who: By tray tightening associate at Final line assembly.

How: Avg. 18 cases / Month.



▪During Metal Battery tray tightening, mounting bolts remains loose.

2. Justify the choice

▪PARETO CHART ANALAYSIS FOR FINAL LINE MARAZZO CONCERNS – Jan 18 to Mar 18

Inference - Tray bolt loose is one of the top concern at final line.

2. Justify the choice (reasons)

Mahindra

PROMISE 2019
Automotive Business

We will…

Double our revenue to ₹53,000 cr (USD 8 bn), while delivering superior financial performance

- Be the 2nd largest OEM in India by volume
- Double the overseas volume with significant presence in 10 markets

By being the most customer centric company

Brand and Customer Satisfaction
- Be among the Top 2 in Customer Satisfaction
- Be among the Top 3 in brand ranking in PV and #1 in UV & SCV
- Have 10 brands with CaPS > 50

Product and Technology
- Launch 18 products
- Be a pioneer in digital customer experience
- Develop one new disruptive business model

Crusade 2.0
- Drive brand preference through Crusade 2.0 of delivering ETR, FTR and DCE

People
- Be the #1 Great Place to Work in the manufacturing sector

Environment
- Reduce carbon and water footprint by 25%

Thus enabling our customers and stakeholders to Rise, as we progress towards being a Top 10 Global Automotive brand.

Mahindra

PROMISE 2019
Nashik Plant 1, Auto Division

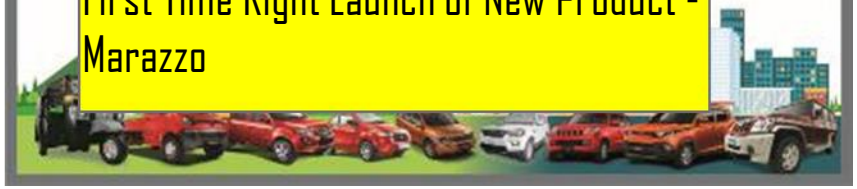
We will…

- Produce 230k vehicles in a year as per market requirement
- Fulfill 100% IO requirement for doubling the overseas volume
- Maintain cost leadership in Auto Sector by reducing revenue cost of 5% YoY
- Improve IQS for Nashik models by 25%, ranking in top 3
- Achieve CaPS > 50 for all Nashik models
- Launch 7 new products on time
- Achieve defined Crusade targets through FTR, ETR and DCE
- Achieve operational excelle[illegible] through digitization
- Achieve No.1 status for em[illegible]ee engagement in the Auto Industry
- Make officers future ready [illegible]plement new technologies coming in the next 5-10 ye[illegible]
- Reduce carbon & water foot[illegible]nt by 25%

we
nd.

Departmental Objective –

First Time Right Launch of New Product - Marazzo



2. Justify the choice

D. COPQ (Cost of Poor Quality) Calculation

Rework cost Estimation	
Description	
Number of pieces (Vehicle) reworked	56
Part cost/piece	0.27
Rework cost/piece(Vehicle) Including (Labor Cost) AVERAGED	18.24
Total scrap cost (Rs.)	198
Total rework cost(Rs.)	3586.15
Total Rejection cost (Rs)	2954.15
(Part Cost + Total rework cost)	
Extrapolated Total rejection cost (Rs.for one year)	Rs.26161.20

2. Justify the choice (reasons)

Problem Criticality (Severity)

- Metal Battery tray bolt loose, if this defect occurs it is resulted into highly customer dissatisfaction because of heavy rattling noise during the driving .

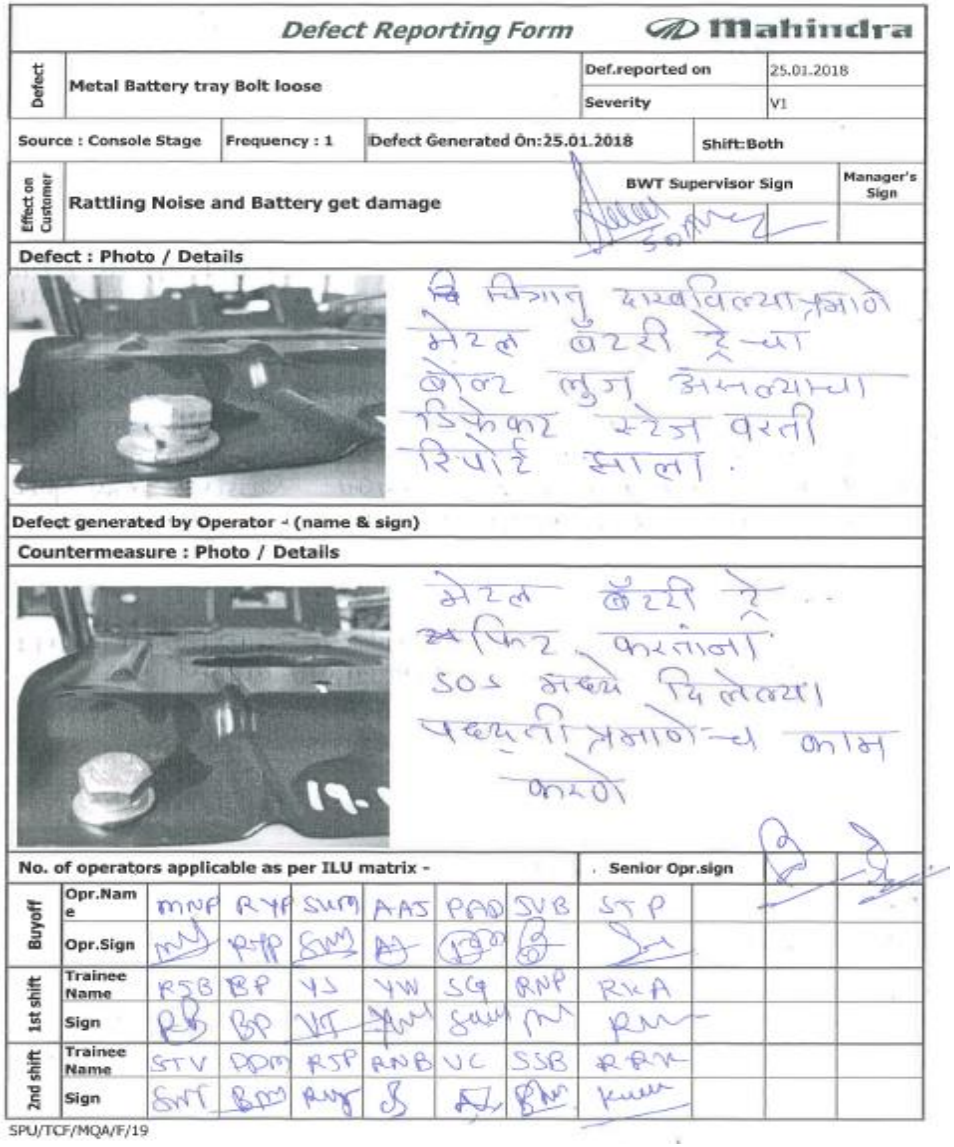
- Metal Battery tray bolts loose contributes share of 2.94% rejection in Shop FRC.

- It creates Unsafe condition to next stage operation.

- Require 12 to 13 mins for rework the one vehicle.

2.b Immediate Containment Action

- Vehicle reworked at FAI rework area for the defect Metal battery tray bolt loose.
- Defect communicated to the stage operator through DRF.



Defect Reporting Form — Mahindra

Defect	Metal Battery tray Bolt loose	Def.reported on	25.01.2018
		Severity	V1
Source : Console Stage	Frequency : 1	Defect Generated On:25.01.2018	Shift:Both
Effect on Customer	Rattling Noise and Battery get damage	BWT Supervisor Sign	Manager's Sign

Defect : Photo / Details

Defect generated by Operator - (name & sign)

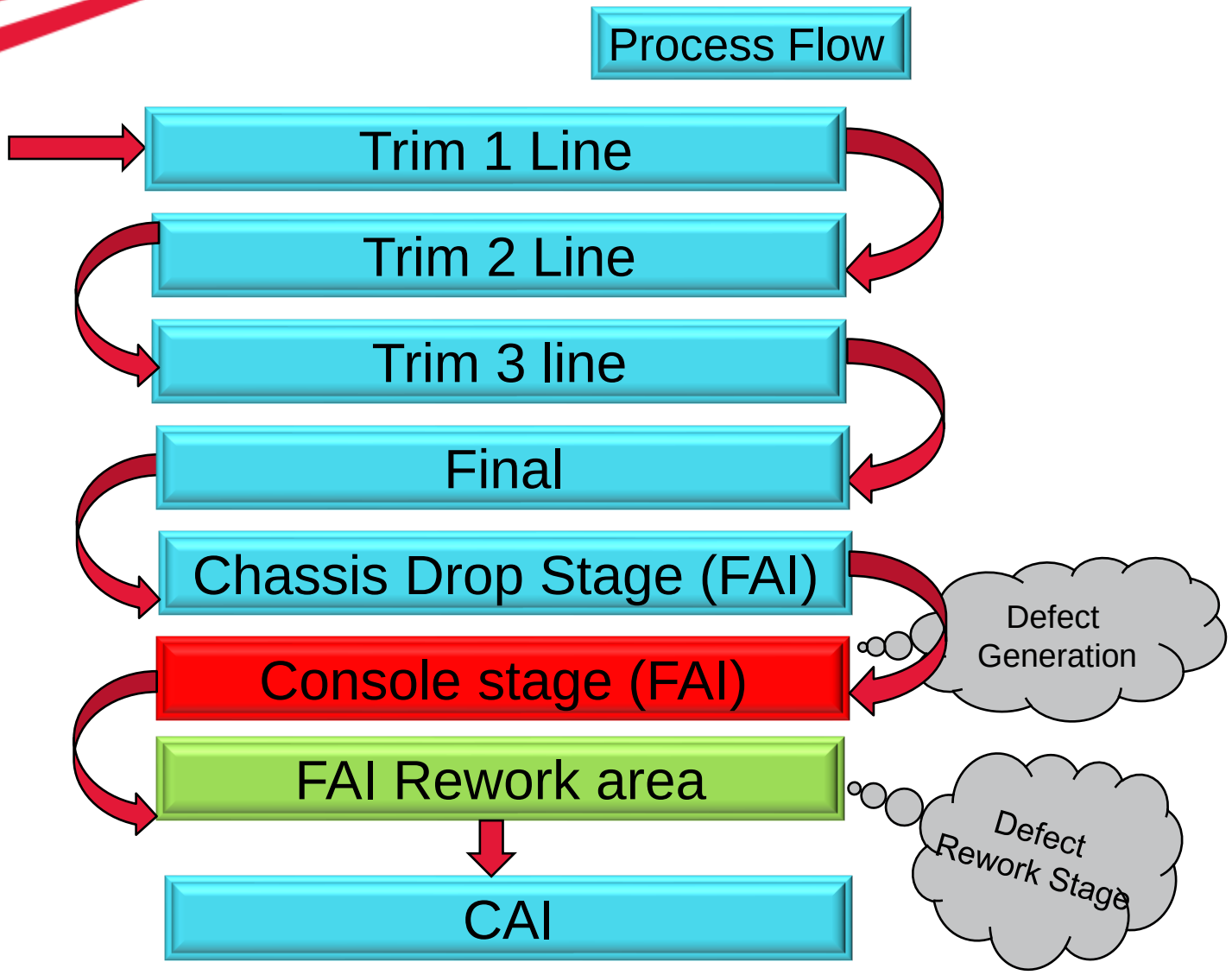
Countermeasure : Photo / Details

No. of operators applicable as per ILU matrix - | Senior Opr.sign

Buyoff	Opr.Name	MNP	RYP	SUM	AAJ	PAD	SVB	STP			
	Opr.Sign										
1st shift	Trainee Name	RSB	BP	VJ	VW	SG	RNP	RKA			
	Sign										
2nd shift	Trainee Name	STV	PDM	RSP	RNB	VC	SSB	RRK			
	Sign										

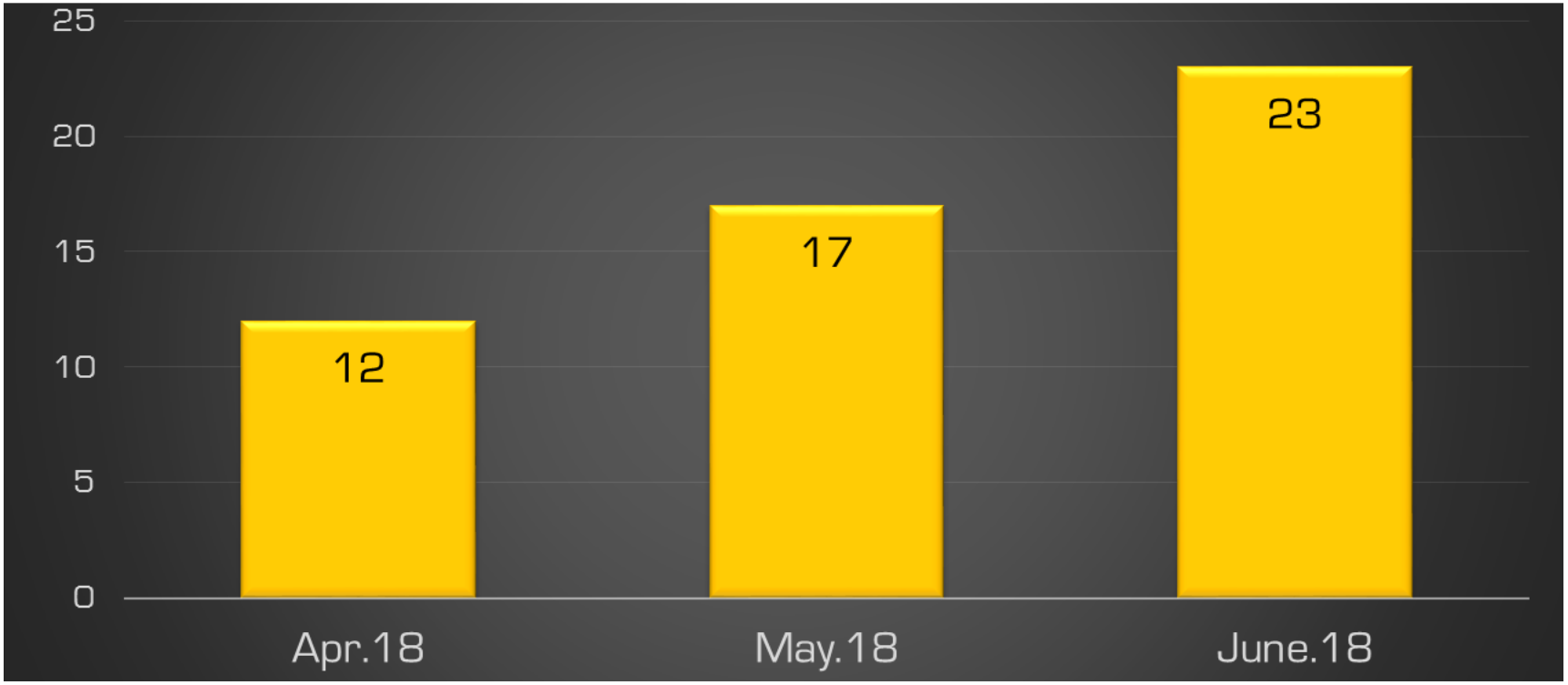
SPU/TCF/MQA/F/19

3. Understand the current situation



3. Understand the current situation

Month wise **Metal Battery Tray bolt loose fitment** concern Trend at final line for Jan-18 to Mar-18



Inference – Metal battery tray bolt loose incidences are increasing month on month.

3. Understand the current situation

Parts used in End Vibration arrestor bracket fitment Process. **(1 Type parts and 1 Type Hardware)**

Metal Battery Tray .

3. Understand the current situation

Parts used

M8 Bolt with Small flange (4 Nos.)

3. Understand the current situation

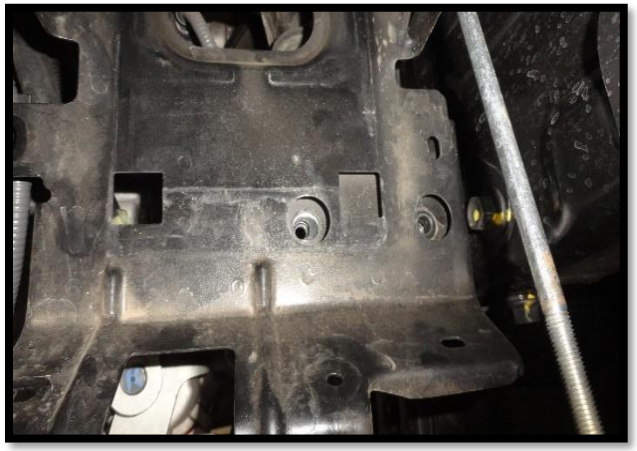
Operation.

Pick UP battery Tray from Line side Material storage trolley

Place the Tray on DAC.

3. Understand the current situation

Operation.

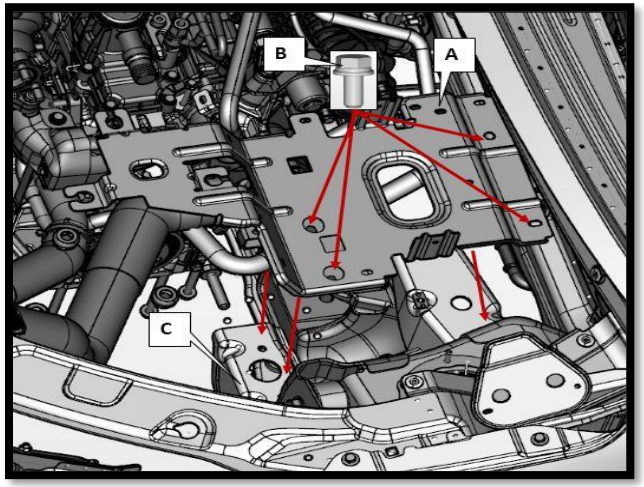


Match the Metal Battery Tray hole with DAC hole by Adjusting the tray.

Pre fit the bolt with 3 to 4 Finger spins .

3. Understand the current situation

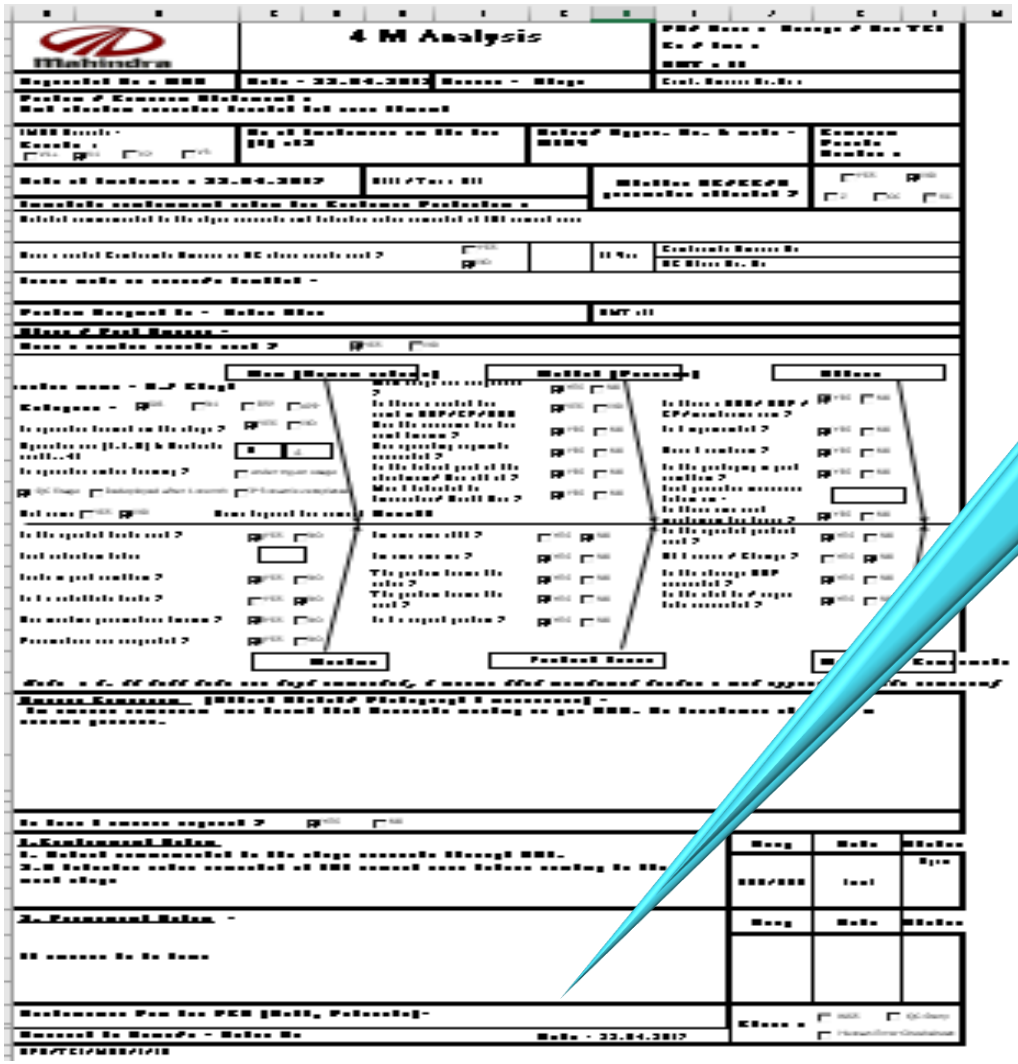
operation.



Location of Four mounting Bolt.

Tightening of Metal Battery Tray.

3. Understand the current situation



Inference from L0
Problem is a repeat problem
L1 analysis required

4M analysis Good and Not so good observations	
MAN	Main step are respected by associate.
Method	Operator aware of key point and check present at buyoff
Others	Control plan adhered on stage
Machine	Tools used are calibrated
Product failure	Problem follows part
Material	Handling and shelf life of part adhered

No Quick Win

3. Understand the current situation

4M Analysis Observation.

- Concern Reported at Console stage.
- **All L4 Operators Working** on Stage.
- Concern Reported in **MARAZZO all model.**
- Concern reported as Metal Battery tray bolt loose at where the bolt does get the full torque.
- **Rework** is done at the rework area- **not possible in tact time at stage**.
- **No access** for **rework** on line.
- **Rework** Takes more than **12 min** time.

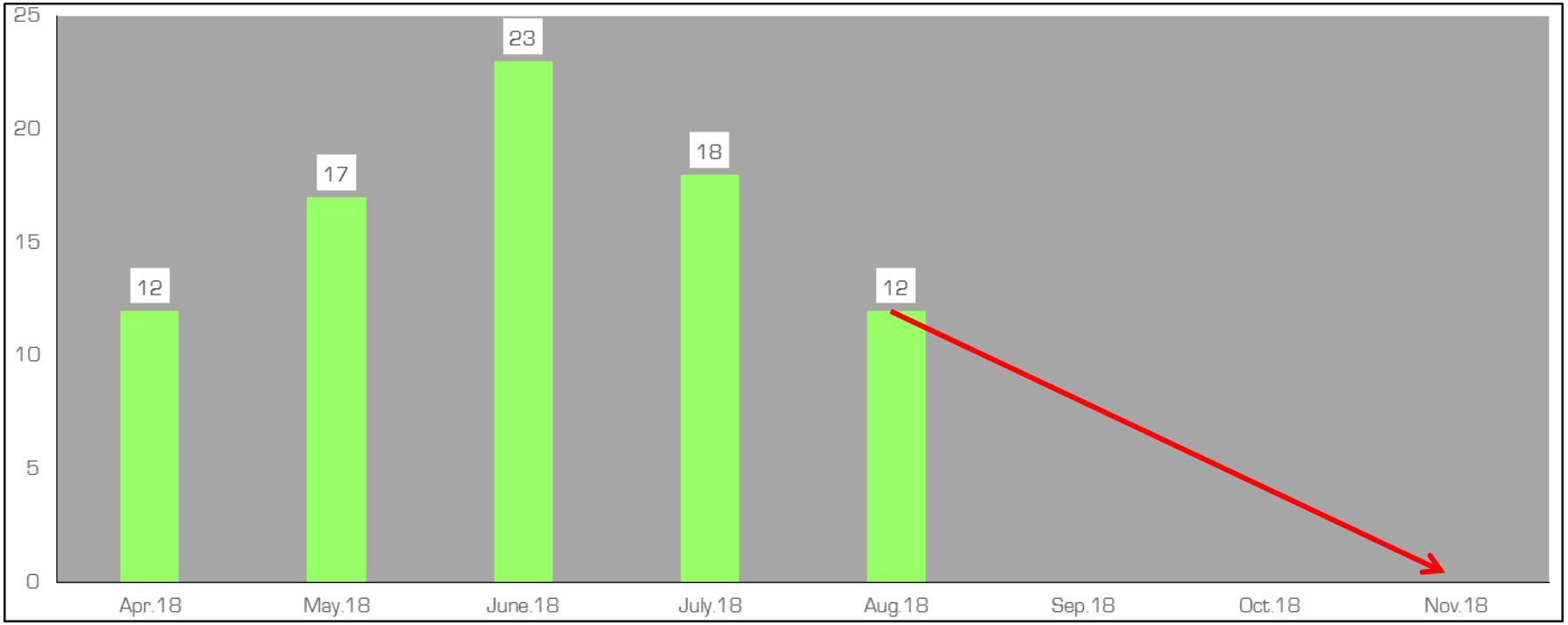
4. Select targets

Present Level:

Average 18 Defects/Month

Target:

Elimination of the defect Metal Battery Tray bolt loose fitment By August 2018



5. Analysis

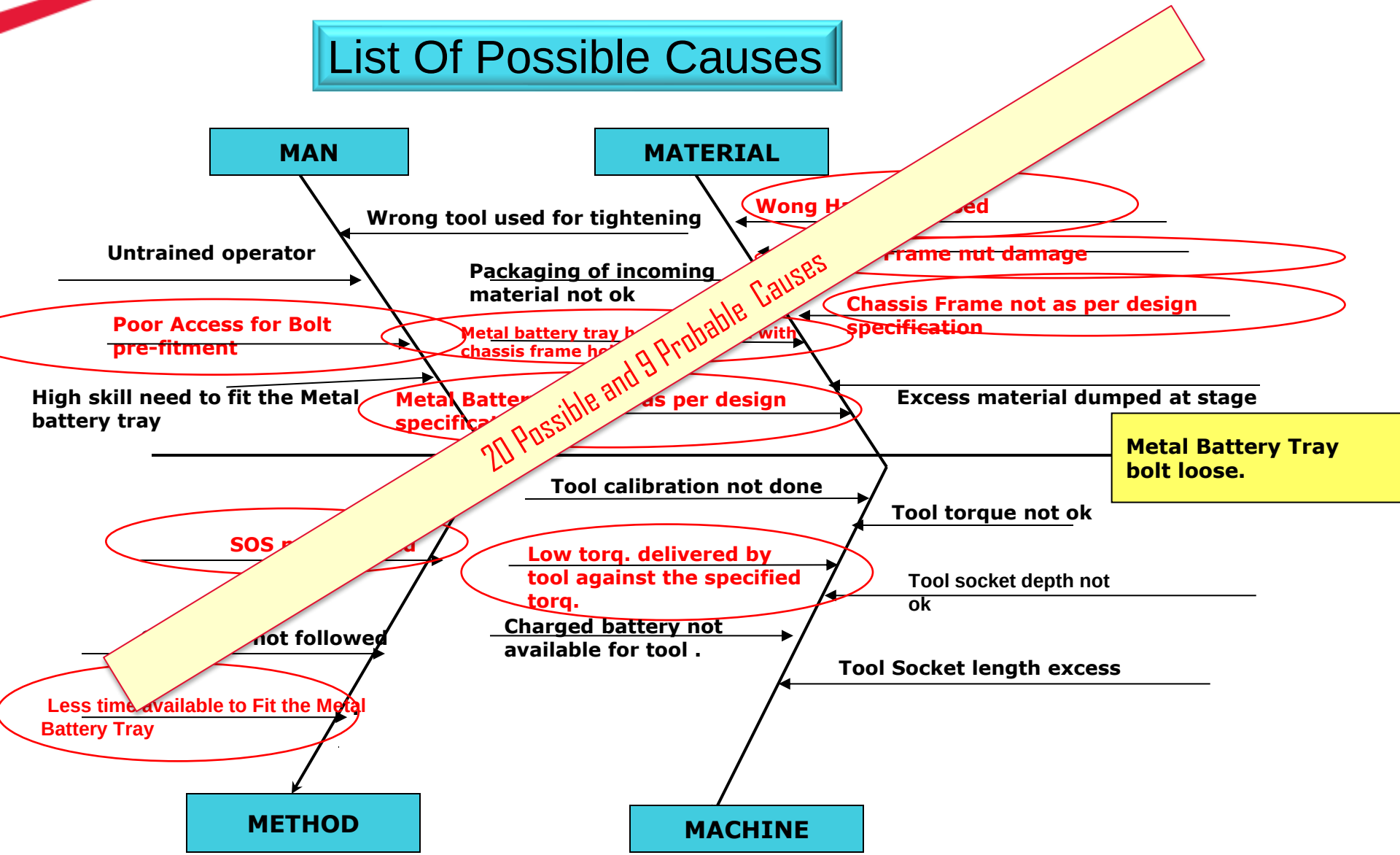
5.1 List of Possible causes: Brainstorming

Sr.No.	Possible cause
1	Untrained operator
2	Wrong tool used for tightening
3	Packaging of incoming material not ok
4	Poor Access Bolt pre fitment
5	High skill need to fit the Metal Battery Tray
6	Metal Battery Tray not as per design specification
7	Excess material dumped at stage
8	Tool torque not ok
9	Wong Hardware used
10	Chassis frame nut damage
11	Chassis Frame not as per design specification
12	Tool socket depth not ok
13	Tool Socket length excess
14	Metal battery tray hole mismatch with chassis frame hole.
15	SOS not followed
16	Tool calibration not done
17	Low torq. delivered by tool against the specified torq.
18	Charged battery not available for tool .
19	Sequence not followed
20	Less time available to Fit the Metal battery tray

Selection Criteria: 0-No impact, 1- less impact , 3-High impact , 5- Significant impact.
Threshold Value for selecting probable cause is 33.

Step 5. Analysis

List Of Possible Causes



5. Analysis

5.1 List of Possible causes: Brainstorming

Sr.No.	Possible cause	Suhas	Kiran	Yogesh	Rajender	Sachin	Total
1	Untrained operator	1	1	3	1	1	7
2	Wrong tool used for tightening	3	1	1	3	1	9
3	Packaging of incoming material not ok	1	1	3	1	1	7
4	Poor Access Bolt pre fitment	9	9	9	3	9	39
5	High skill need to fit the Metal Battery Tray	1	3	1	1	3	9
6	Metal Battery Tray not as per design specification	9	9	3	9	9	39
7	Excess material dumped at stage	3	1	1	3	1	9
8	Tool torque not ok	0	1	1	1	1	4
9	Wong Hardware used	9	9	9	9	9	45
10	Chassis frame nut damage	9	9	9	9	9	45
11	Chassis frame not as per design specification	9	9	9	9	9	45
12	Tool socket depth not ok	0	0	1	1	1	3
13	Tool Socket length excess	0	3	1	1	3	8
14	Metal battery tray hole mismatch with chassis frame hole.	9	9	9	9	9	45
15	SOS not followed	9	9	3	9	3	33
16	Tool calibration not done	3	1	1	1	3	9
17	Low torq. delivered by tool against the specified torq.	9	9	9	9	9	45
18	Charged battery not available for tool .	1	1	3	1	1	7
19	Sequence not followed	3	1	3	1	3	11
20	Less time available to Fit the Metal battery tray	9	9	9	9	9	45

Selection Criteria: 0-No impact, 1- less impact , 3-High impact , 5- Significant impact.
Threshold Value for selecting probable cause is 33.

Step 5. Analysis

5.1 List of Probable causes: Brainstorming

List of causes not considered as probable causes and reasons for not considering

SR No.	Not considered as cause	Reason
1	Untrained operator	Trained operator deployed on stage
2	Wrong tool used for tightening	Similar looking other tool is not used on same stage
3	Packaging of incoming material not ok	Packaging is in good condition
4	High skill need to fit the Metal Battery Tray	All associate are L3 and L4 level
5	Excess material dumped at stage	Two bin system is available on stage.
6	Tool torque not ok	Tool torque found ok.
7	Tool socket depth not ok	Any depth socket can be used.
8	Tool Socket length excess	Any length socket can be used.
9	Tool calibration not done	Tool calibration done as per schedule.
10	Charged battery not available for tool .	Sufficient charged batteries available on stage
11	Sequence not followed	Associate working as per SOS

5. Analysis

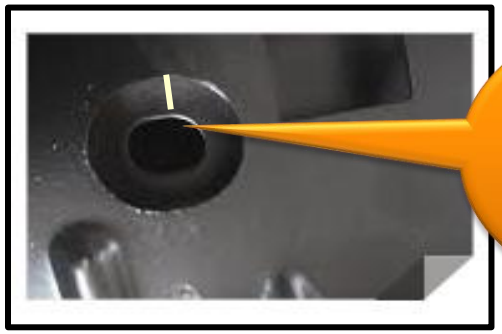
5.1 List of Probable causes: Brainstorming

Sr.No.	Possible cause
1	Poor Access Bolt pre fitment
2	Metal Battery Tray not as per design specification
3	Wong Hardware used
4	Chassis frame nut damage
5	Chassis Frame not as per design specification
6	Metal battery tray hole mismatch with chassis frame hole.
7	SOS not followed
8	Low torq. delivered by tool against the specified torq.
9	Less time available to Fit the Metal battery tray

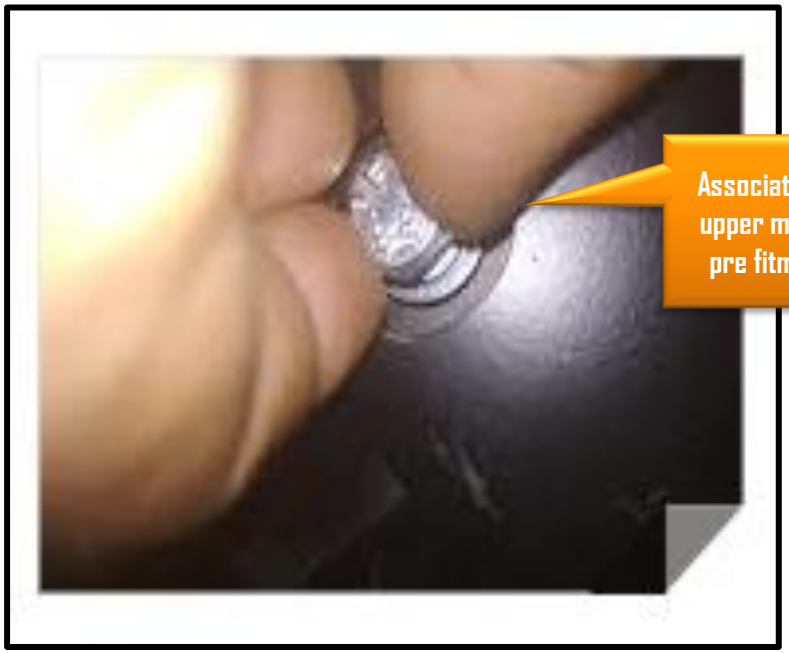
TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
1	Poor Access Bolt pre fitment .	Visual observations	CFT member observe the stage for 2 hrs. Observations are as below... 1.In every vehicle associate facing poor accessibility issue.	**Hypothesis valid**



12 MM Depth between two metal sheets (Upper and lower sheet



Associate finger fouling to the upper metal sheet during bolt pre fitment in 100 % vehicle

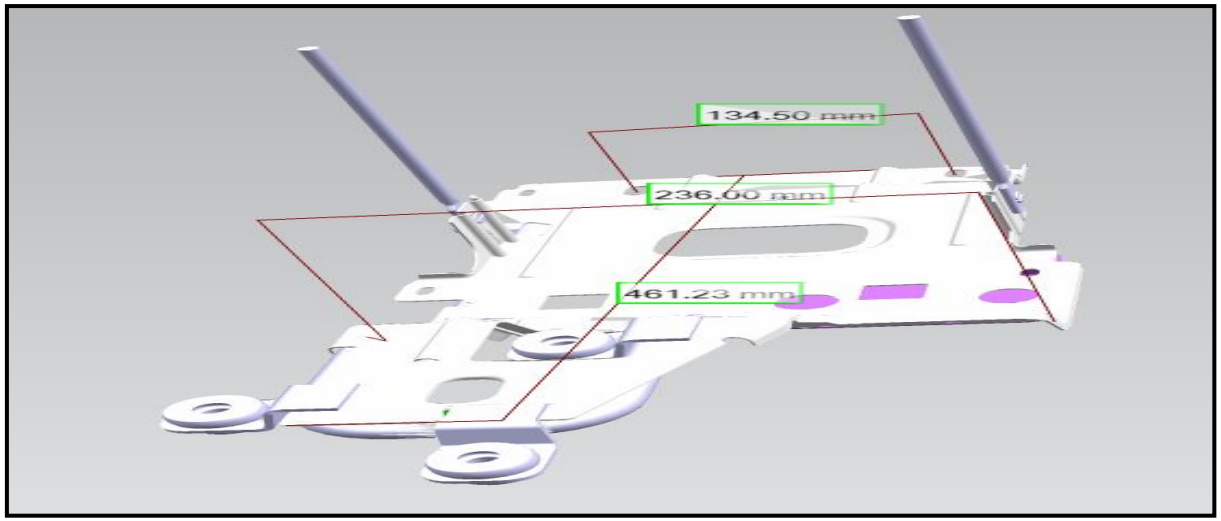
As shown in Picture there are Metal sheet (Upper and Lower) With 12 MM depth between two metal sheets.

Associate finger fouling to Upper metal sheet which creates difficulty in Bolt pre fitment in 100% Vehicle.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
2	Metal Battery Tray not as per design specification.	Vernier Caliper	5 Nos. End vibration bracket checked found within design specification.	Hypothesis Invalid



Sr.No.	Distance between Two Slots		Length along X Axis		Width along Y Axis	
	Specification	Actul	Specification	Actul	Specification	Actul
1	134mm +/-0.2	134.1mm	461mm +/-1	461mm	236mm +/-0.1	236.1mm
2	134mm +/-0.2	134.2mm	461mm +/-1	461mm	236mm +/-0.1	236.1mm
3	134mm +/-0.2	134.1mm	461mm +/-1	461mm	236mm +/-0.1	236.1mm
4	134mm +/-0.2	134.1mm	461mm +/-1	460mm	236mm +/-0.1	236.0mm
5	134mm +/-0.2	134.1mm	461mm +/-1	460mm	236mm +/-0.1	236.1mm

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
3	Wong Hardware used.	Visual	CFT done GEMBA visit on Stage. Similar looking hardware is not used for same or other model. So CFT conclude that there is No chance of wrong hardware usage on the stage.	Hypothesis Invalid

Similar Looking Hardware not available on same stage

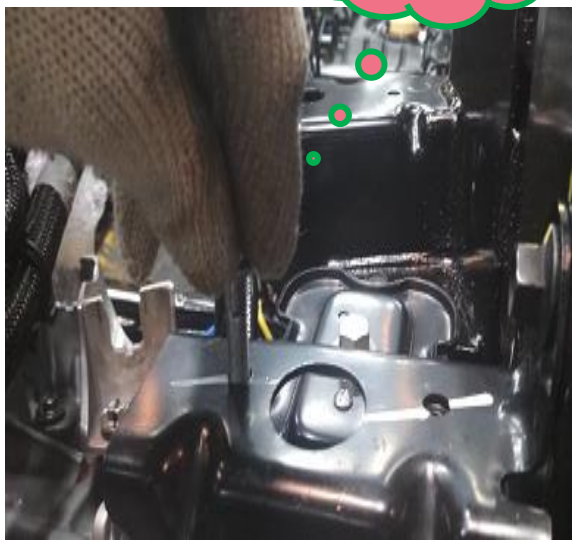
TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
4	Chassis frame nut damage.	By using GO gauge	20 DAC checked for Bumper panel and reinforcement structure nut damage. All DAC found ok	Hypothesis Invalid

Chassis frame nut damage.	
Sr. No	**Observation**
1	Found OK
2	Found OK
3	Found OK
4	Found OK
5	Found OK
6	Found OK
7	Found OK
8	Found OK
9	Found OK
10	Found OK
11	Found OK
12	Found OK
13	Found OK
14	Found OK
15	Found OK
16	Found OK
17	Found OK
18	Found OK
19	Found OK
20	Found OK

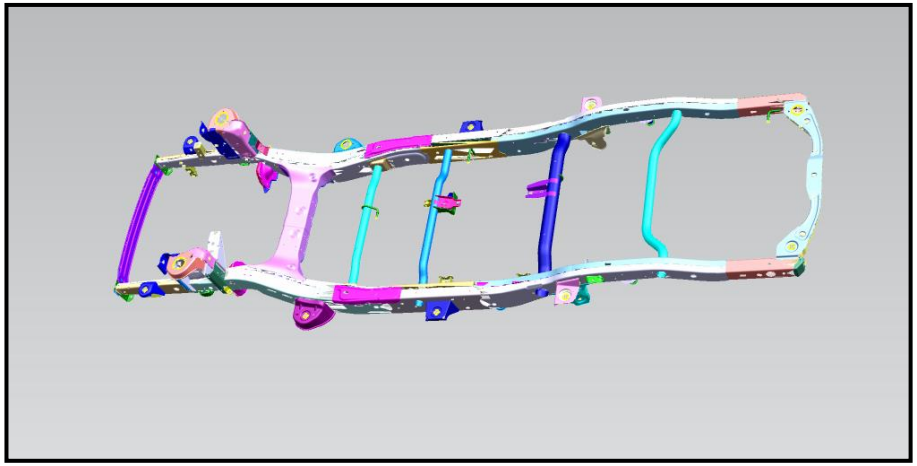
Nut checking by Go gauge

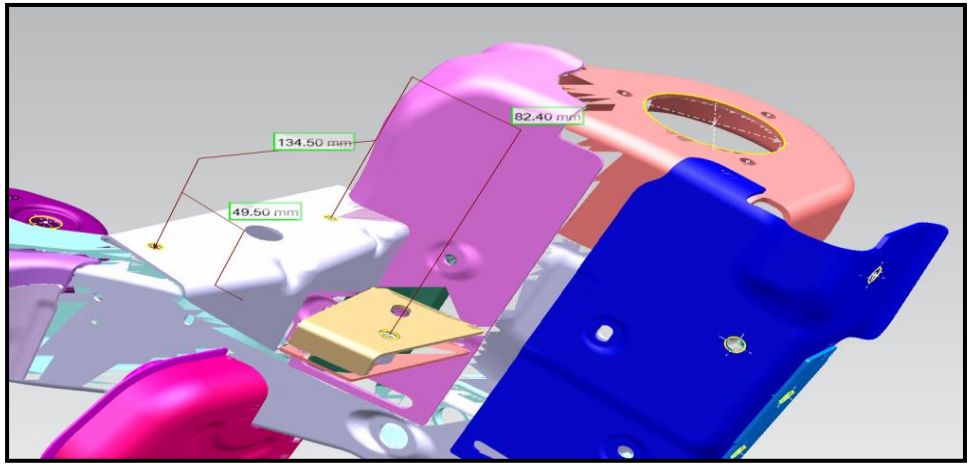


TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
5	Chassis Frame not as per design specification.	Vernier Caliper/Measuring Tape	5 Nos. Reinforcement structure checked found within design specification.	Hypothesis Invalid





Sr.No.	Distance between Two Slots		Distance between Frt & RR Slots	
	Specification	Actul	Specification	Actul
1	134mm +/-0.2	134.0mm	82mm +/-0.1	82.1mm
2	134mm +/-0.2	134.2mm	82mm +/-0.1	82.1mm
3	134mm +/-0.2	134.0mm	82mm +/-0.1	81.9mm
4	134mm +/-0.2	134.1mm	82mm +/-0.1	81.9mm
5	134mm +/-0.2	134.0mm	82mm +/-0.1	82.1mm

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
6	Metal battery tray hole mismatch with chassis frame hole.	Visual	30 Vehicles observed by CFT for the concern. Out of 30 vehicles, in 12 vehicles LH side Rear hole of battery tray not matching with chassis frame hole.	**Hypothesis valid**

Metal battery tray hole mismatch with chassis frame hole.

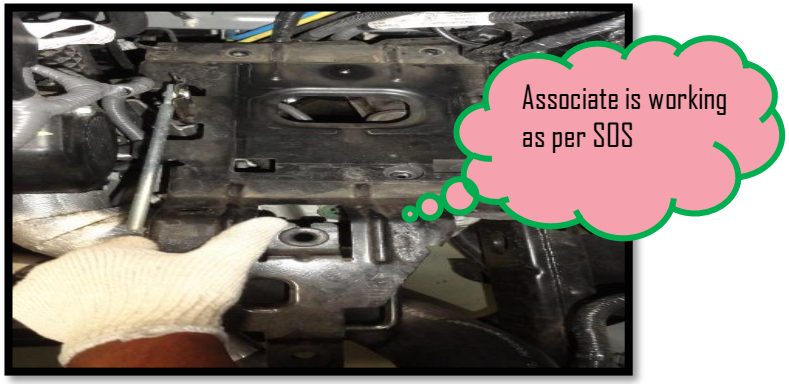
Sr. No	Observation
1	Found OK
2	Found Not OK
3	Found OK
4	Found Not OK
5	Found Not OK
6	Found Not OK
7	Found OK
8	Found Not OK
9	Found OK
10	Found Not OK
11	Found Not OK
12	Found OK
13	Found OK
14	Found Not OK
15	Found OK
16	Found OK
17	Found OK
18	Found OK
19	Found OK
20	Found Not OK
21	Found OK
22	Found OK
23	Found OK
24	Found OK

Metal battery tray hole mismatch with Chassis frame hole

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
7	SOS not followed	Visual	58 cycles observed. Associate assembled and tighten the part as per defined SOS.	Hypothesis Invalid



TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
8	Low torq. delivered by tool against the specified torq.	By digital torq. wrench	32 times tool torque checked on ACTA. Tool delivered Torque within specification.	Hypothesis Invalid

Low torq. delivered by tool against the specified torq.

Sr.No.	Spcified Torque	Torque observed
1	25 NM	24.1 NM
2	25 NM	24.8 NM
3	25 NM	24.1 NM
4	25 NM	24.6 NM
5	25 NM	24.8 NM
6	25 NM	25.1 NM
7	25 NM	24.9 NM
8	25 NM	24.1 NM
9	25 NM	25.2 NM
10	25 NM	24.6 NM
11	25 NM	24.8 NM
12	25 NM	25.1 NM
13	25 NM	24.9 NM
14	25 NM	24.1 NM
15	25 NM	25.2 NM
16	25 NM	24.8 NM
17	25 NM	24.1 NM
18	25 NM	24.6 NM
19	25 NM	24.8 NM
20	25 NM	25.1 NM
21	25 NM	24.1 NM
22	25 NM	24.8 NM
23	25 NM	24.6 NM
24	25 NM	24.1 NM
25	25 NM	24.9 NM
26	25 NM	24.8 NM
27	25 NM	24.1 NM
28	25 NM	24.8 NM
29	25 NM	24.1 NM
30	25 NM	24.6 NM
31	25 NM	24.8 NM
32	25 NM	24.1 NM

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
9	Less time available to Fit the Metal Battery Tray.	MOST Study and Stop Watch study	12 Vehicles data checked for available TACT. All vehicles work completed in available TACT.	Hypothesis Invalid

Less Time Available for Battery Tray Fitment	
Sr.No.	Observation
1	Found OK
2	Found OK
3	Found OK
4	Found OK
5	Found OK
6	Found OK
7	Found OK
8	Found OK
9	Found OK
10	Found OK
11	Found OK
12	Found OK

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
1	Poor Access Bolt pre fitment .	Visual observations	ember observe the stage for 2 hrs. bservations are as below... 1.In every vehicle associate facing poor accessibility issue.	Hypothesis valid
2	Metal battery tray hole mismatch with chassis frame hole.	Visual	30 Vehicles observed by CFT for the concern. Out of 30 vehicles, in 12 vehicles LH side Rear hole of battery tray not matching with chassis frame hole.	Hypothesis valid

2 Valid Causes

5.3 Identify the root causes:

WHY - WHY Analysis

No	Valid Probable Cause	WHY	WHY	WHY	ACTION / COUNTERMEASURE
1	Poor Access Bolt pre fitment .	Difficult to align the bolt inside mounting Hole	Fingers touches to the side wall of Metal battery tray HOLE due to two metal sheet.	Bolt length is less & No dog point available.	1. Bolt length to be increase 2.Dog point to be provide.
2	Metal battery tray hole mismatch with chassis frame hole.	Difficult to align the Metal battery tray hole with chassis frame hole.	LH Rear side hole metal battery tray hole not get match with chassis frame hole.	LH rear side Hole does not accommodate the chassis frame variation due to close tolerance	1. LH Rear side hole to be modify as a slot to accommodate the chassis hole variation in " X " axis.

6. Implement the corrective measures

6.1 Actions Decided

No	Root Cause	Action plan	Target date	Responsibility	Status
1	Bolt length is less also No dog pint available.	1. Bolt length to be increase 2.Dog point to be provide.	09.08.2018	Varun Parakh	Closed
2	LH rear side Hole does not accommodate the chassis frame variation due to close tolerance	1. LH Rear side hole to be modify as a slot to accommodate the chassis hole variation in " X " axis	12.08.2018	Varun Parakh	Closed

6. Implement the corrective measures

No	Root Cause	Action plan	Target date	Responsibility	Status
1	Bolt length is less also No dog pint available.	1. Bolt length to be increase 2.Dog point to be provide.	09.08.2018	Varun Parakh	Closed

Before

After

Engineering NOTICE

Mahindra & Mahindra Ltd.
Product Development

"Confidential Report"

Page No: 1

Mahindra Rise

Engineering Notice Release: U321-EN-A003-000055
AD Product Development
Mahindra & Mahindra LTD(A.S)

Created Date 25-Jul-2018
Initiated Date 03-Aug-2018

Engineering Notice No : U321-EN-A003-000055
Released Date : 21-Aug-2018 17:43
Reference Change Req No : DCR041275
Location : AD-CHENNAI
Implementation Location : M&M-Nasik
Engineering Notice Description : FOAM ADDITION IN RAT MESH AND BATTERY TRAY STIFFENING
Reason : FOAM ADDITION IN RAT MESH AND BATTERY TRAY STIFFENING
Quantity : -(Regular)
Reason Code : New Variant introduction through EN
Priority : Mandatory
Note : EN RELEASE FORFOAM ADDITION IN RAT MESH AND BATTERY TRAY STIFFENIN
Affected Models : ALL,
Is EN Re Released? :
Parent EN NO :
Life Cycle State : Implemented
Owner Name : Parakh Varun (23078746)

Engineer : Parakh Varun (23078746)
Approved By : 301332-JAYASHANKAR V
Authorized By : U321-HEAD_301184-NAIR SAJITH

Legends

Add New : A Part/Assy that is getting released through ECN for the very first time
Modified : A Part/Assy that is already released through an ECN and has undergone some changes
Added with Modification : It is very similar to "Modified" status except that it is being added newly into a different Parent
Existing : A Part/Assy that is already being used is carried forward as it is to some other parent
Deleted : A Part/Assy that is being used is removed from a Parent
Reinstated : Reintroduction of a Part/Assy that was used Previously in M&M but was discontinued

ICA :
O - New part number can be used in place of old part no. & vice versa;
I - New part number can be used in place of old part no. but not vice versa;
X - Old part number can be used in place of new part no. but not vice versa;
N - No ICA between new and old part numbers;
NA - Interchangeability as set of parts (to use new part no. we have to use more than one new part no);
NA - Not Applicable;

Mahindra Rise

Engineering Notice Release: U321-EN-A003-0000
AD Product Development
Mahindra & Mahindra LTD(A.S)

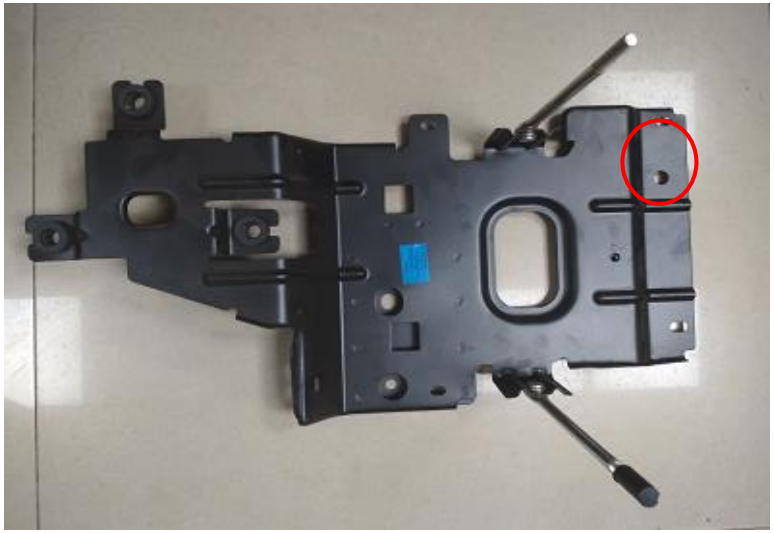
Created Date 25-Jul-2018
Initiated Date 03-Aug-2018

Mounting Bolt changed

Part Number	Rev	Nomenclature	Status	ICA	Qty	Replaced/ Deleted Part	Parent Part		Responsibility	Remarks	Discontinued
0000AU30011LA	*D/015*	*U321 AIR INTAKE ASSEMBLY*	*Modified*	*NA*	*1*		*U321-EBOM*		*NIMGAONKAR SAMEER (208505)*		*No*
SF0103080	A/001	BOLT COMBI LP M8X1.25X21X8.8	Add Existing	I	4	SF0103010	1401BAA0013LA		NIMGAONKAR SAMEER (208505)		No
1401BAA03590N	D/016	BRACKET ASSY BATTERY TRAY	Modified	NA	1		1401BAA0013LA		NIMGAONKAR SAMEER (208505)		No
SF0103010	A/001	BOLT COMBI LP M8X1.25X21X8.8XZN	Deleted	NA	4		1401BAA0013LA		NIMGAONKAR SAMEER (208505)		No
0313ABF00150N	C/024	DIESEL AIR INTAKE SYSTEM LAYOUT	Modified	NA	1		0000AU30011LA		PISAT_JITENDRA (23070155)		No
1401BAA0013LA	D/016	LAYOUT BRKT ASSY BAT TRY WI FSTN	Modified	NA	1		0000AU30011LA		NIMGAONKAR SAMEER (208505)		No
1401BAA03570N	D/014	BRACKET BATTERY TRAY AIRBOX	Modified	NA	1		1401BAA03590N		NIMGAONKAR SAMEER (208505)		No
1401BAA03580N	B/006	BRKT TUBE TRAY AIRBOX	Modified	NA	1		1401BAA03590N		NIMGAONKAR SAMEER (208505)		No
0313ABF00160N	B/014	DUCT ASSY DIRTY SIDE	Modified	NA	1		0313ABF00150N		PISAT_JITENDRA (23070155)		No
0313ABF00120N	B/025	HOSE ASSY CLEAN SIDE INTAKE	Modified	NA	1		0313ABF00150N		PISAT_JITENDRA (23070155)		No
0313ABF00140N	B/015	HOSE INTERMEDIATE PIPE TO TURBO	Modified	NA	1		0313ABF00120N		PISAT_JITENDRA (23070155)		No
0313EAM00030N	A/002	FOAM DUCT DIRTY	Add New	NA	1		0313ABF00160N		PISAT_JITENDRA (23070155)		No

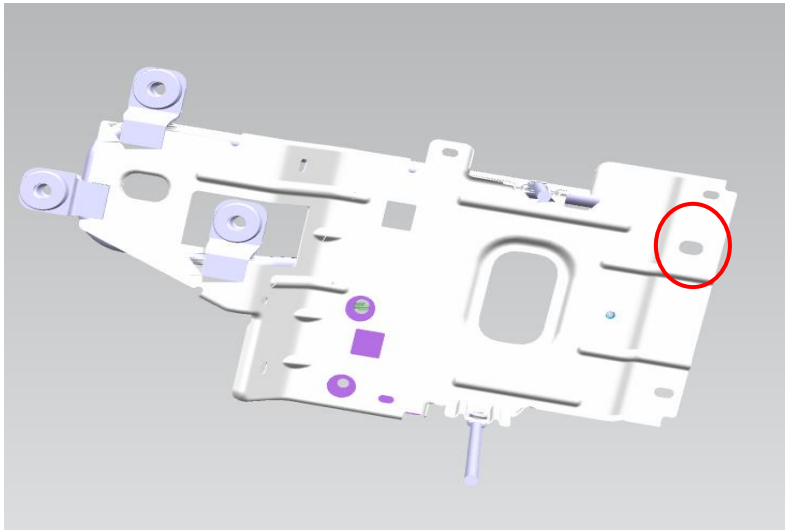
6. Implement the corrective measures

No	Root Cause	Action plan	Target date	Responsibility	Status
1	LH rear side Hole does not accommodate the chassis frame variation due to close tolerance	1. LH Rear side hole to be modify as a slot to accommodate the chassis hole variation in " X " axis	12.08.2018	Varun Parakh	Closed



Round Hole

Before



Hole modified in Slot

After

Engineering NOTICE

Mahindra & Mahindra Ltd.
Product Development

"Confidential Report"

Page No 1

Mahindra Rise

Engineering Notice Release: U321-EN-A003-000075
AD Product Development
Mahindra & Mahindra LTD(A.S)

Created Date 28-Sep-2018

Initiated Date

Engineering Notice No : U321-EN-A003-000075

Released Date :

Reference Change Req No :

Location : AD-CHENNAI

Implementation Location : M&M-Nasik

Engineering Notice Description : SLOT MODIFICATION IN BATTERY TRAY

Slot modification in battery tray

Reason : Slot modication in battery tray

Quantity : -(Regular)

Reason Code : Manufacturing / Process Improvements

Priority : Mandatory

Note : EN RELEASE FOR SLOT MODIFICATION IN BATTERY TRAY

Affected Models : all,

Is EN Re Released? :

Parent EN NO :

Life Cycle State : Working

Owner Name : Parakh Varun (23078746)

Engineer : Parakh Varun (23078746)

Approved By : 301332-JAYASHANKAR V

Authorized By : U321-HEAD_207094-GAUBA MANOJ

Legends

Add New : A Part/Assy that is getting released through ECN for the very first time

Modified : A Part/Assy that is already released through an ECN and has undergone some changes

Added with Modification : It is very similar to "Modified" status except that it is being added newly into a different Parent

Existing : A Part/Assy that is already being used is carried forward as it is to some other parent

Deleted : A Part/Assy that is being used is removed from a Parent

Reinstated : Reintroduction of a Part/Assy that was used Previously in M&M but was discontinued

ICA :
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X - No ICA between new and old part numbers;
Y - Interchangeability as set of parts (to use new part no, we have to use more than one new part no);
NA - Not Applicable;

7. Check & confirm the results & effects

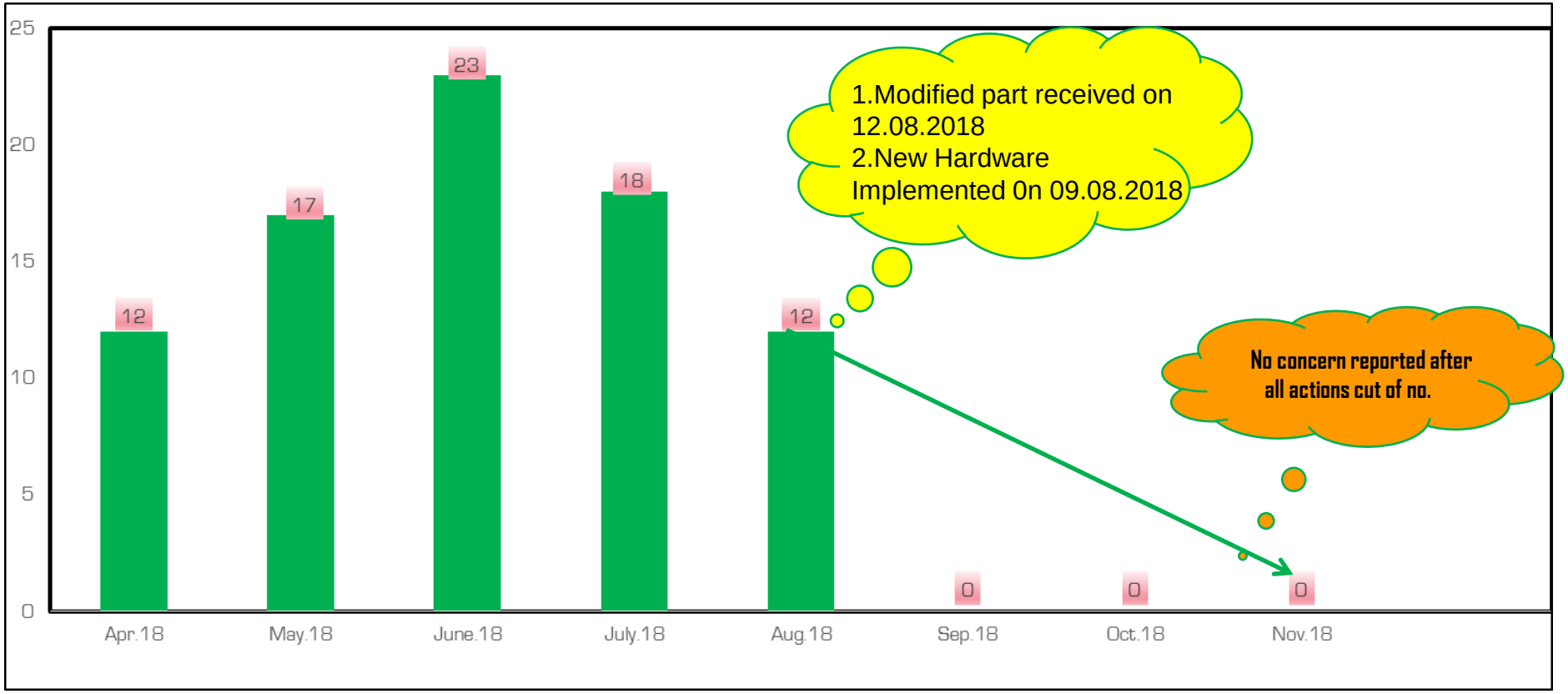
7.1 Verify Results:

S N	Action Taken	Testing Method	Results Before Action	Results After Action
1	1. Bolt length to be increased 2.Dog point to be provided	No Poor Accessibility for Bolt pre-fitment.	In 100% vehicle associate face the poor Accessibility issue for Bolt pre fitment	Zero poor accessibility issue for bolt pre fitment .
2	LH Rear side hole to be modify as a slot to accommodate the chassis hole variation in " X " axis	Visual	Daily 3 to 4 cases per shift.	Zero cases after modification done in part.

7. Check & confirm the results & effects

7.1 Verify Results: Defect Trend

X Axis- Month
Y Axis-Roof liner ding at rear side in Scorpio vehicle concern numbers.



Inference - Concern is eliminated after all 3 actions implemented.

7. Check & confirm the results & effects

7.2 Full Scale Implementation plan:

S N	Action Taken	Target date	Responsibility	Status
1	1. Bolt length to be increased 2.Dog point to be provided	09.08.2018	Varun Parakh	Closed
2	LH Rear side hole to be modify as a slot to accommodate the chassis hole variation in " X " axis	12.08.2018	Varun Parakh	Closed

7. Check & confirm the results & effects

7.2 Full Scale Implementation plan:

S N	Action Taken	Target date	Responsibility	Status
3	Part Drawing	12.08.2018	Varun Parakh	Closed
4	SOS Updation.	25.08.2018	Suhas Aher	Closed
5	PFMEA updation.	25.08.2018	Suhas Aher	Closed

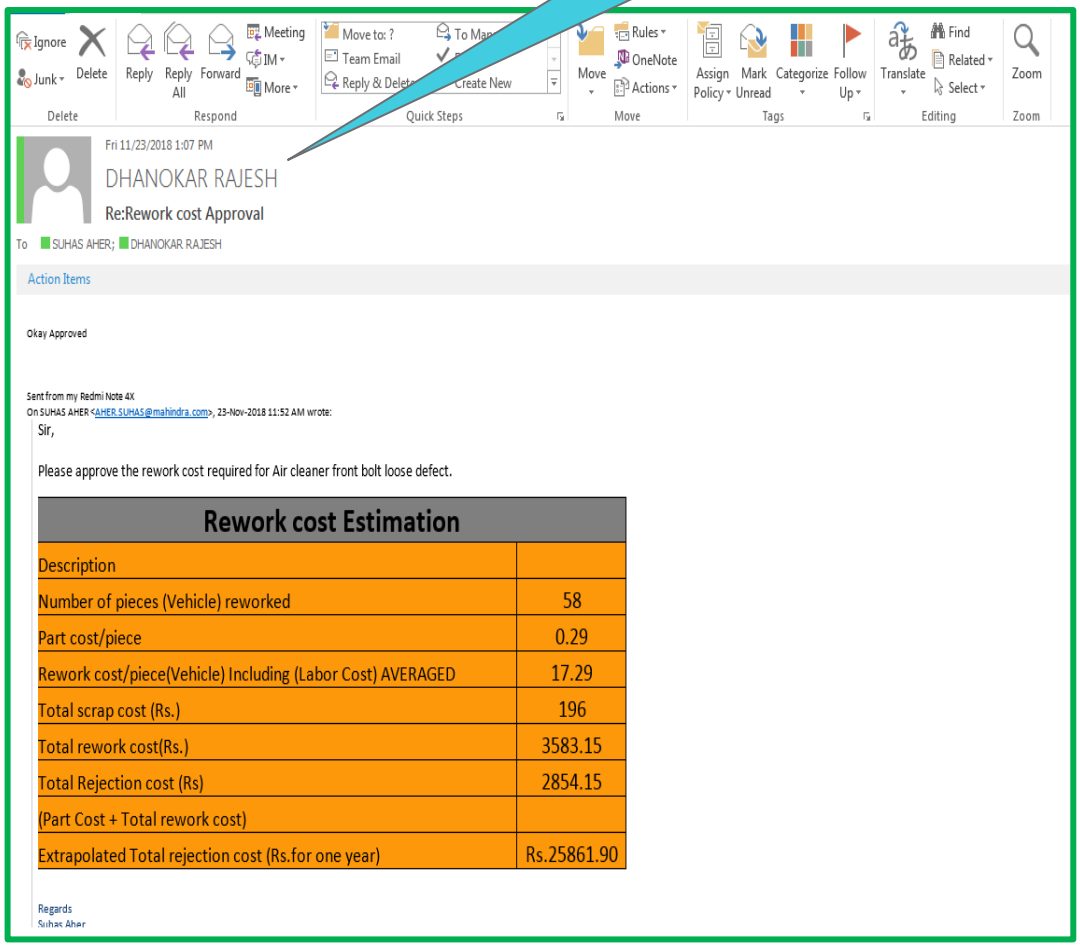
Check & confirm the results & effects (Cont..)

7.3 Tangible & Intangible benefits:

Certified by Industrial Engineer

Tangible Benefits -

1) DPT reduction by 356
2) COPQ (Material cost+ Labour cost+ Rework cost) =(Apprx. Rs.25861.90 Rs. Per annum)is Saved



Fri 11/23/2018 1:07 PM

DHANOKAR RAJESH

Re:Rework cost Approval

To SUHAS AHER; DHANOKAR RAJESH

Action Items

Okay Approved

Sent from my Redmi Note 4X

On SUHAS AHER <AHER.SUHAS@mahindra.com>, 23-Nov-2018 11:52 AM wrote:

Sir,

Please approve the rework cost required for Air cleaner front bolt loose defect.

Rework cost Estimation	
Description	
Number of pieces (Vehicle) reworked	58
Part cost/piece	0.29
Rework cost/piece(Vehicle) Including (Labor Cost) AVERAGED	17.29
Total scrap cost (Rs.)	196
Total rework cost(Rs.)	3583.15
Total Rejection cost (Rs)	2854.15
(Part Cost + Total rework cost)	
Extrapolated Total rejection cost (Rs.for one year)	Rs.25861.90

Regards

Suhas Aher

Check & confirm the results & effects (Cont..)

7.3 Tangible & Intangible benefits:

In tangible Benefits

1) Associate fatigue reduced
2) Improved Customer Satisfaction.

8. Standardization

8.1 Updation of documents: e.g. Part Drawing,EN,SOS,PFMEA,Check sheet.

SN	Document	Target date	Responsibility	Status
1	Part Drawing.	12.08.2018	Varun Parakh	Closed
2	PFMEA updation.	25.08.2018	Suhas Aher	Closed
3	SOS Updation.	25.08.2018	Suhas Aher	Closed

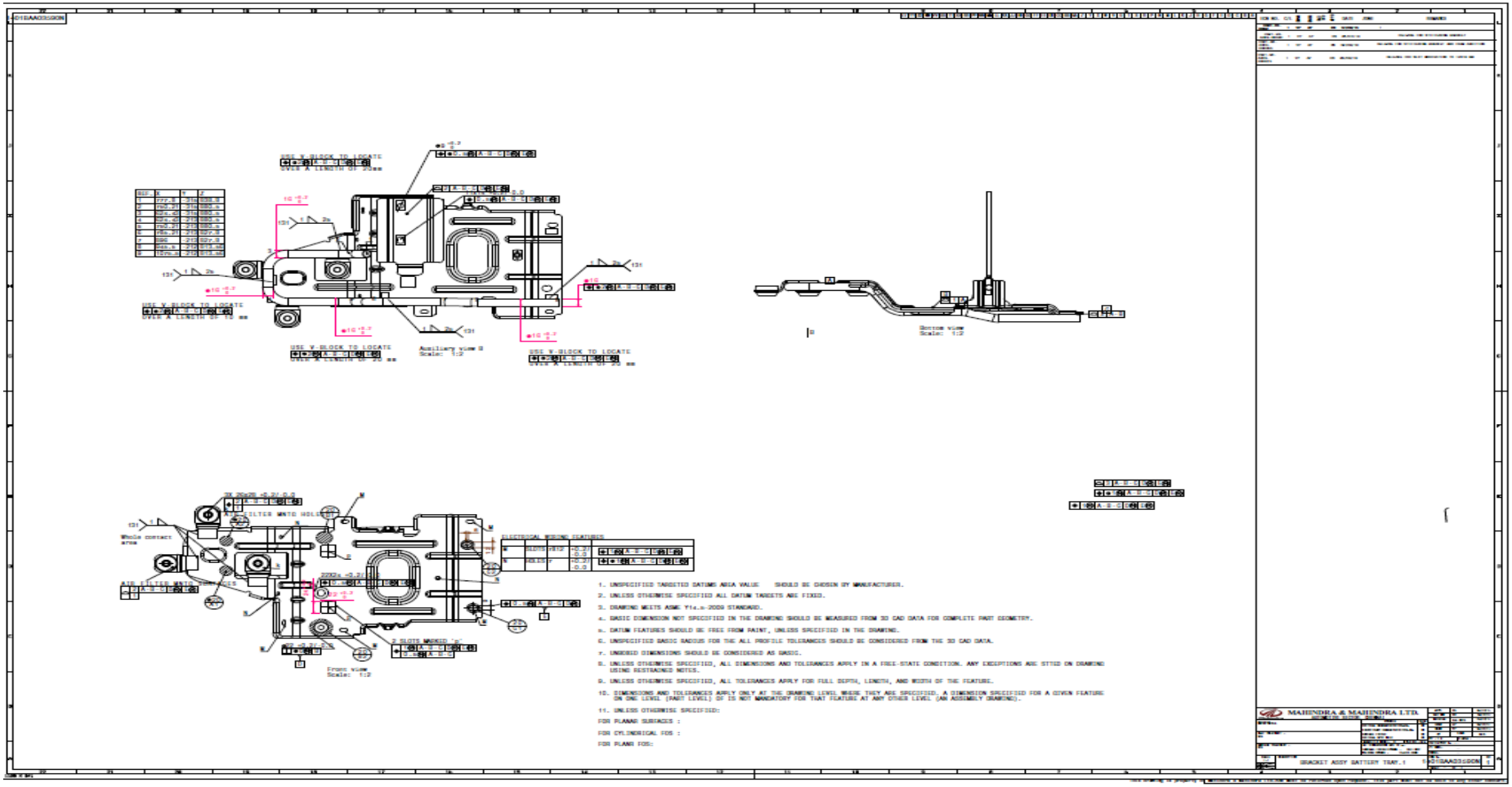
8.2 Training:

SN	Training Identified	Target date	Responsibility	Status
1.	On job training given to all stage associate and rework associate for part modification.	18.08.2018	Suhas Aher	Closed

8. Standardization

8.1 Updation of documents: e.g. Part Drawing,EN,SOS,PFMEA,Check sheet.

PART DRAWING



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ENGINEERING NOTICE

Mahindra & Mahindra Ltd.
Product Development

"Confidential Report"

Page No 1

Mahindra Rise

Engineering Notice Release: U321-EN-A003-000075
AD Product Development
Mahindra & Mahindra LTD(A.S)

Created Date 28-Sep-2018
Initiated Date

Engineering Notice No : U321-EN-A003-000075

Released Date :

Reference Change Req No :

Location : AD-CHENNAI

Implementation Location : M&M-Nasik

Engineering Notice Description : SLOT MODIFICATION IN BATTERY TRAY

Slot modification in battery tray

Reason : Slot modication in battery tray

Quantity : -(Regular)

Reason Code : Manufacturing / Process Improvements

Priority : Mandatory

Note : EN RELEASE FOR SLOT MODIFICATION IN BATTERY TRAY

Affected Models : all,

Is EN Re Released? :

Parent EN NO :

Life Cycle State : Working

Owner Name : Parakh Varun (23078746)

Engineer : Parakh Varun (23078746)

Approved By : 301332-JAYASHANKAR V

Authorized By : U321-HEAD_207094-GAUBA MANOJ

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ENGINEERING NOTICE

Engineering Notice Release: U321-EN-A003-000055
AD Product Development
Mahindra & Mahindra LTD(A.S)

Created Date 25-Jul-2018
Initiated Date 03-Aug-2018

Engineering Notice No : U321-EN-A003-000055
Reference Change Req No : DCR041275
Location : AD-CHENNAI
Implementation Location : M&M-Nasik
Engineering Notice Description : FOAM ADDITION IN RAT MESH AND BATTERY TRAY STIFFENING
Reason : FOAM ADDITION IN RAT MESH AND BATTERY TRAY STIFFENING
Reason Code : New Variant Introduction through EN
Note : EN RELEASE FORFOAM ADDITION IN RAT MESH AND BATTERY TRAY STIFFENIN
Affected Models : ALL,
Is EN Re Released? :
Life Cycle State : Implemented

Released Date : 21-Aug-2018 17:43
Quantity : -(Regular)
Priority : Mandatory
Parent EN NO :
Owner Name : Parakh Varun (23078746)

Engineer : Parakh Varun (23078746)
Approved By : 301332-JAYASHANKAR V
Authorized By : U321-HEAD_301184-NAIR SAJITH

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N - Interchangeability as set of parts (to use new part no. we have to use more than one new part no);
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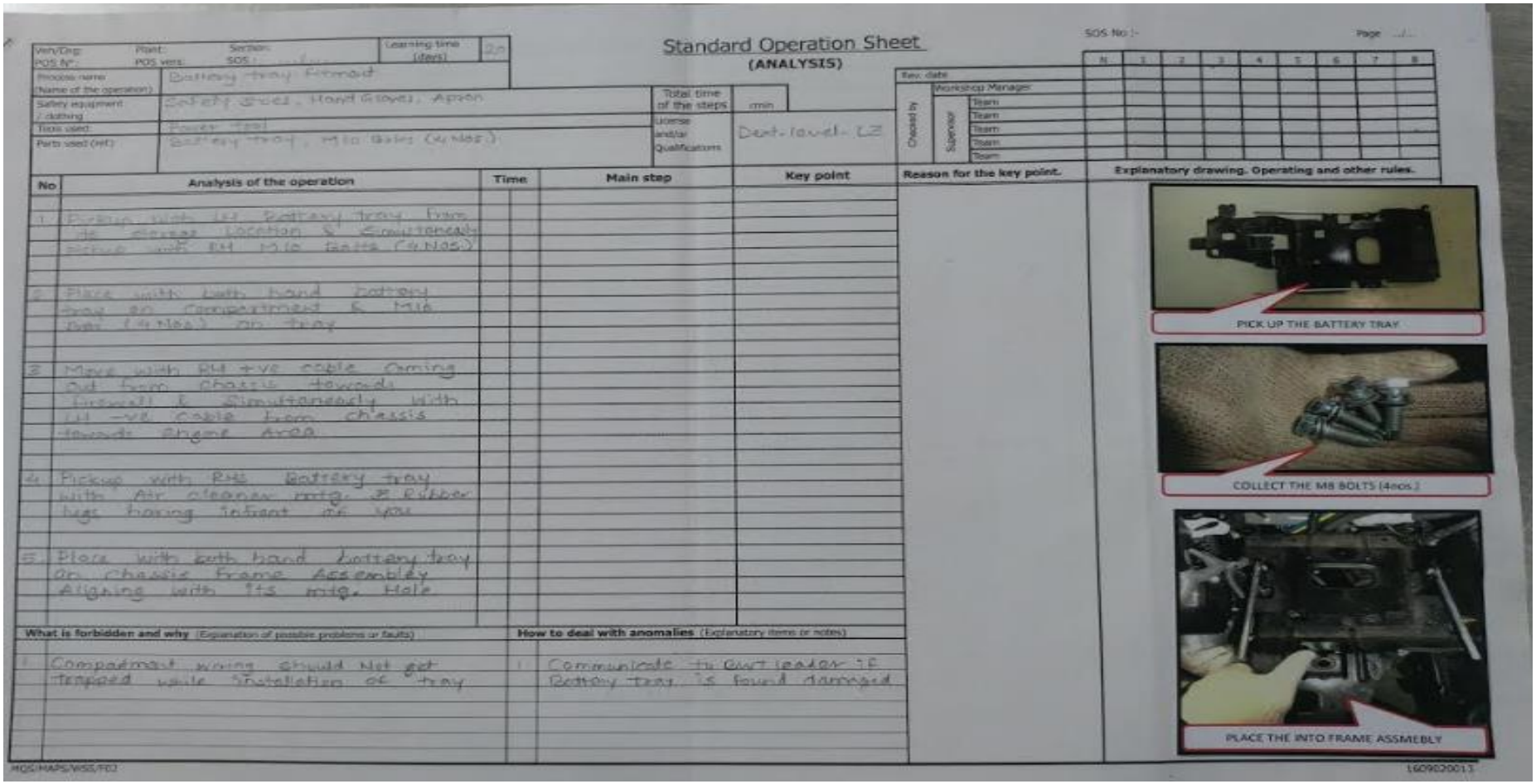
Mounting Bolt changed

Part Number	Rev	Nomenclature	Status	ICA	Qty	Replaced/ Deleted Part	Parent Part		Responsibility	Remarks	Discontinued
0000AU30011LA	*D/015*	*U321 AIR INTAKE ASSEMBLY*	*Modified*	*NA*	*1*		*U321-EBOM*		*NIMGAONKAR SAMEER (208505)*		*No*
SF0103080	A/001	BOLT COMBI LP M8X1.25X21X8.8	Add Existing	I	4	SF0103010	1401BAA0013LA		NIMGAONKAR SAMEER (208505)		No
1401BAA03590N	D/016	BRACKET ASSY BATTERY TRAY	Modified	NA	1		1401BAA0013LA		NIMGAONKAR SAMEER (208505)		No
SF0103010	A/001	BOLT COMBI LP M8X1.25X21X8.8XZN	Deleted	NA	4		1401BAA0013LA		NIMGAONKAR SAMEER (208505)		No
0313ABF00150N	C/024	DIESEL AIR INTAKE SYSTEM LAYOUT	Modified	NA	1		0000AU30011LA		PISAT_JITENDRA (23070155)		No
1401BAA0013LA	D/016	LAYOUT BRKT ASSY BAT TRY WI FSTN	Modified	NA	1		0000AU30011LA		NIMGAONKAR SAMEER (208505)		No
1401BAA03570N	D/014	BRACKET BATTERY TRAY AIRBOX	Modified	NA	1		1401BAA03590N		NIMGAONKAR SAMEER (208505)		No
1401BAA03580N	B/006	BRKT TUBE TRAY AIRBOX	Modified	NA	1		1401BAA03590N		NIMGAONKAR SAMEER (208505)		No
0313ABF00160N	B/014	DUCT ASSY DIRTY SIDE	Modified	NA	1		0313ABF00150N		PISAT_JITENDRA (23070155)		No
0313ABF00120N	B/025	HOSE ASSY CLEAN SIDE INTAKE	Modified	NA	1		0313ABF00150N		PISAT_JITENDRA (23070155)		No
0313ABF00140N	B/015	HOSE INTERMEDIATE PIPE TO TURBO	Modified	NA	1		0313ABF00120N		PISAT_JITENDRA (23070155)		No
0313EAM00030N	A/002	FOAM DUCT DIRTY	Add New	NA	1		0313ABF00160N		PISAT_JITENDRA (23070155)		No

8. Standardization

8.1 Updation of documents: e.g. Part Drawing,EN,SOS,PFMEA,Check sheet.

SOS



Standard Operation Sheet
(ANALYSIS)

Ver/Dig: Plant: Section: Learning time (days): 20
POS By: POS vers: SOS:
Process name (Name of the operation): Battery tray Fitment
Safety equipment / clothing: Safety shoes, Hand Gloves, Apron
Tools used: Power tool
Parts used (ref): Battery tray, M10 Bolts (4 Nos.)
Total time of the steps: min
License and/or Qualifications: Dent-level-L2

SOS No.:- Page ../..
Key date
Checked by: Workshop Manager: Supervisor: Team Leader / Team / Team / Team

No	Analysis of the operation	Time	Main step	Key point	Reason for the key point.	Explanatory drawing. Operating and other rules.
1	Pickup with left Battery tray from its storage location & simultaneously pickup with RH M10 Bolts (4 Nos.)					
2	Place with both hand battery tray on compartment & M10 Bolt (4 Nos.) on tray					
3	Move with RH +ve cable coming out from chassis towards firewall & simultaneously with LH -ve cable from chassis towards engine area					
4	Pickup with RHS Battery tray with Air cleaner mtg. & Rubber legs having infront of you					
5	Place with both hand battery tray on chassis frame assembly aligning with its mtg. Hole					

What is forbidden and why (Explanation of possible problems or faults)	How to deal with anomalies (Explanatory items or notes)
1 Compartment wiring should not get trapped while installation of tray	1 Communicate to our leader if Battery tray is found damaged

MQS/MAPS/WSS/F02 160902001.3

8. Standardization

8.1 Updation of documents: e.g. Part Drawing,EN,SOS,PFMEA,Check sheet.

PFMEA

Mahindra Rise

MAHINDRA & MAHINDRA
AUTOMOTIVE AND FARM SECTOR

Format Name: PFMEA | Format No: MPDS/PC/F14 | Rev. No.00 | Rev. Date:08/04/2016

Initiated By: CME

Plant : Nashik Plant
Part Name:
Latest Change Letter:NA
PMF Shop (Sample YES
Nodal/ Cell (Plant

Model (Sample
Part Name (Process)
Rear Bumper vibration
Rqts No.:
Rqts Model:
Rear Bumper vibration arrester mtg bracket
R/C Mark & No.:NA

Mahindra & Mahindra Ltd.

Process Potential Failure Mode & Effect Analysis.

Reviewed by : Sohan Ahir
Approved by : Pravin Patil
Rev. No :
Rev. Date :
Doc No.: PFMEA/PPR/SCOPE/FI MAS
FMEA Date (Orig) :22.08.16

Cross Functional Team
Sachin Biraar
Pitambar Shelke
Baharakah Auhate
Vikrant Ghate
Mandar Joshi

Process Step / Function	Requirement	Potential Failure Mode	Potential Effect(s) of Failure	SEV	Class	Potential Cause(s)/Mechanism(s) of Failure	Current Process Controls Prevention	OCC	Current Process Controls Detection	DET	RPN	Recommended Action(s)	Responsibility & Target Completion Date	Action Results: Action Taken	SEV	OCC	DET	RPN
Rear Bumper vibration arrester mtg bracket fitment	Rear Bumper vibration arrester mtg bracket fitment on Bumper & Chassis	One or more bolts not assembled	Mfg & Assy: Rework on next stage End User: Rattle noise and noticed by: TB & customer	4		Operator forgot to tighten 3 bolts.	Training to operator	2	Check at tightening stage Buyoff stage Squak & Rattle ANOVA-O/R/F1	7	56							
	Tighten 3 bolts to specified torque.	Bolts under torque	Mfg & Assy: Rework on next stage End User: Rattle noise and noticed by: TB & customer	4		Improper power tool calibration	Calibration as per frequency decided in control plan	4	Sticker for due date for calibration on tool	7	112							
			Mfg & Assy: Rework on next stage End User: Rattle noise and noticed by: TB & customer	4		Operator did not confirm the gap between bolt & check arm	Process is mentioned in SOS Training to operator as per SOS	2	Check at buyoff	7	56							
						Use of wrong power tool	Provision of dedicated power tool for this application	2	Identification sticker is provided on the	3	24							
				4		Due to improper accessibility		10		7	280	Weld nut from Stay bracket to be shifted on L Bracket Stud to be provided For mounting 2 bolts on Bumper side Stay bracket side bolt to be change to stud to 8 direction	Sachin Patil					
		Hex nut/bolt overtorqued	Mfg & Assy: Rework on next stage End User: Rattle noise and noticed by: TB & customer	4		Improper power tool calibration	Calibration as per frequency decided in control plan	2	Sticker for due date for calibration on tool	3	24							
						Use of wrong power tool	Provision of dedicated power tool for this application	2	Identification sticker is provided on the	3	24							

Rev No.:	Rev Date	Revision Summary	Reason for Revision
01	22.09.16	W109 project	W109 project

8. Standardization

8.3 Additional Audit / Check

6	Check point added at console stage	25.08.2018	Suhas Aher.	Closed

U321 CONSOLE STAGE - Check sheet

Sr. No.	SC CC	Description	What To Check	Sev.	How To Check	Why To check	How To Mark	Source of complaint
1		All EGR pipe connection.	1.Routing as per SOS. 2.All Ends fully insert in respective port. 3.No Kink. 4.Connections as per SOS.		1.Visual 2.Marking	1.Pipe get Kink,cut,Damage and		
2	SC	Pneumatic cable routing on Pulley						
3		Pulley mounted four bus.						
4		Metal Pulley tray	1. No loose fitment	V1	1.Touch and feel	1.Pulley get damage. 2.Rattle noise		Visual
5		Body Harness cable routing.	1.Mounting clips fully lock on metal pulley tray.[3 Nos.]	V1	1.Visual and marking	1.Cable get cut		Relative Marking
6	SC	ECU Bracket.	1.Bracket fully lock in ECU also secondary lock completely lock .[4Nos.] 2.Bracket bracket clip fully lock on Dash bracket.[4 Locations] 3.Bracket- No damage	V1	1.Pull Push and Marking. 2.Pull Push and Marking. 3.Visual	1.Vehicle inspectable. 2.Wire get cut. 3.Vehicle inspectable.		Relative Marking
7		Engine and Engine room wire loom interconnection.	1.Completly lock. 2.Completly mounted on engine bracket.	V1	1.Pull Push and marking. 2.Visual	1.Vehicle will not start. 2.Bracket get damage over the period		Relative Marking
8		Radiator oil.	1.On per side.[LHRH] 2.Top peak pin fully lock.[1 Nos each side] 3.Bottom leg completely insert in mounted hole.[1 Nos. each side]	V2	1.Visual. 2.Touch and feel and Marking 3.Touch and feel	1.Poor sealing effect. 2. Poor sealing effect and rattling noise. 3.Poor sealing effect and rattling noise.		Relative Marking
9		Front End cross member to Lower Energy absorber joint. [1No Screw each side LHRH]	1.No loose fitment	V1	1.Touch and feel and Marking	1.Rattling Noise.		Relative Marking
10		Upper Air Deflector to Lower Energy absorber joint. [1 No big size flange nut each side]	1.No loose fitment 2.Big flange type nut.	V1	1.Visual 2.Touch and feel and marking	1.Rattling Noise. 2.Not dig nut and part get damage.		Relative Marking

4		Metal Battery tray	1. No loose fitment	V1	1.Touch and feel	1.Battery get damage. 2.Rattle noise		Visual

9. Summaries & Plan future actions

9.1 Conclusion:

Metal battery tray bolt loose fitment is eliminated in Marazzo after

- Mounting Bolt length increase.
- Dog Point provided on mounting Bolt.
- Slot provided on at LH Rear side mounting area i.p.o of hole

9.2 Horizontal Deployment:

All U322 and U325 model From VP gateway.

9.3 Lessons Learnt:

Things Gone Wrong :

Initially focus were only given of manufacturing process.

Things Gone Right :

With systematic study of all causes & CFT approach help to resolve the concern.

Thank You.....